

Analytical Report

MGMT-6216
Predictive Analytics

Clustering Socio-Economic Profiles of Countries

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Part 1: Data Preparation and Exploration

Loading and Inspecting the Dataset

The dataset contains 167 countries and 10 variables, covering economic indicators (income, GDP per capita, exports, imports, inflation) and health indicators (child mortality, life expectancy, fertility rate, health expenditure).

```
> dim(df)
[1] 167  10
> head(df)
# A tibble: 6 × 10
  country   child_mort exports health imports income inflation life_expec total_fer  gdpp
  <chr>          <dbl>   <dbl>  <dbl>   <dbl>  <dbl>     <dbl>      <dbl>     <dbl> <dbl>
1 Afghanis…       90.2    10     7.58    44.9   1610      9.44        56.2      5.82   553
2 Albania         16.6    28     6.55    48.6   9930      4.49        76.3      1.65  4090
3 Algeria         27.3    38.4   4.17    31.4  12900     16.1         76.5      2.89  4460
4 Angola         119      62.3   2.85    42.9   5900     22.4         60.1      6.16  3530
5 Antigua …       10.3    45.5   6.03    58.9  19100      1.44        76.8      2.13 12200
6 Argentina       14.5    18.9   8.1     16    18700     20.9         75.8      2.37 10300
```

Figure 1: Dataset dimensions and preview

Checking for Missing Values and Duplicates

```
> print("Rows with missing values:")
[1] "Rows with missing values:"
> print(missing_rows)
# A tibble: 0 × 10
# ℹ 10 variables: country <chr>, child_mort <dbl>, exports <dbl>, health <dbl>,
#   imports <dbl>, income <dbl>, inflation <dbl>, life_expec <dbl>, total_fer <dbl>,
#   gdpp <dbl>
```

Figure 2: Missing value check

```
> print(paste("Total duplicate rows:", duplicate_count))
[1] "Total duplicate rows: 0"
```

Figure 3: Duplicate row check

Summary Statistics and Visualizations

```
> summary(df)
   country           child_mort       exports           health          imports
 Length:167         Min.   :  2.60   Min.   :  0.109   Min.   : 1.810   Min.   :  0.0659
 Class :character   1st Qu.:  8.25   1st Qu.: 23.800   1st Qu.: 4.920   1st Qu.: 30.2000
 Mode  :character   Median : 19.30   Median : 35.000   Median : 6.320   Median : 43.3000
                    Mean   : 38.27   Mean   : 41.109   Mean   : 6.816   Mean   : 46.8902
                    3rd Qu.: 62.10   3rd Qu.: 51.350   3rd Qu.: 8.600   3rd Qu.: 58.7500
                    Max.   :208.00   Max.   :200.000   Max.   :17.900   Max.   :174.0000
     income         inflation        life_expec      total_fer           gdpp
 Min.   :   609   Min.   : -4.210   Min.   :32.10   Min.   :1.150   Min.   :   231
 1st Qu.:  3355   1st Qu.:  1.810   1st Qu.:65.30   1st Qu.:1.795   1st Qu.:  1330
 Median :  9960   Median :  5.390   Median :73.10   Median :2.410   Median :  4660
 Mean   : 17145   Mean   :  7.782   Mean   :70.56   Mean   :2.948   Mean   : 12964
 3rd Qu.: 22800   3rd Qu.: 10.750   3rd Qu.:76.80   3rd Qu.:3.880   3rd Qu.: 14050
 Max.   :125000   Max.   :104.000   Max.   :82.80   Max.   :7.490   Max.   :105000
```

Figure 4: Summary statistics of all variables

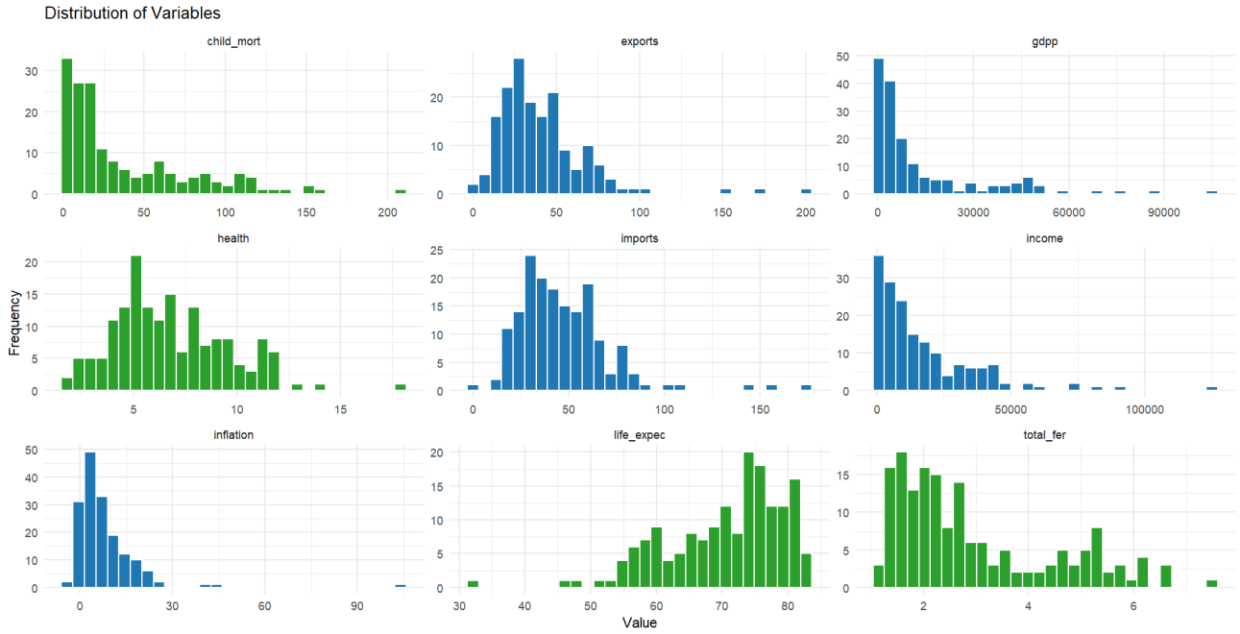


Figure 5: Histograms showing distribution of variables

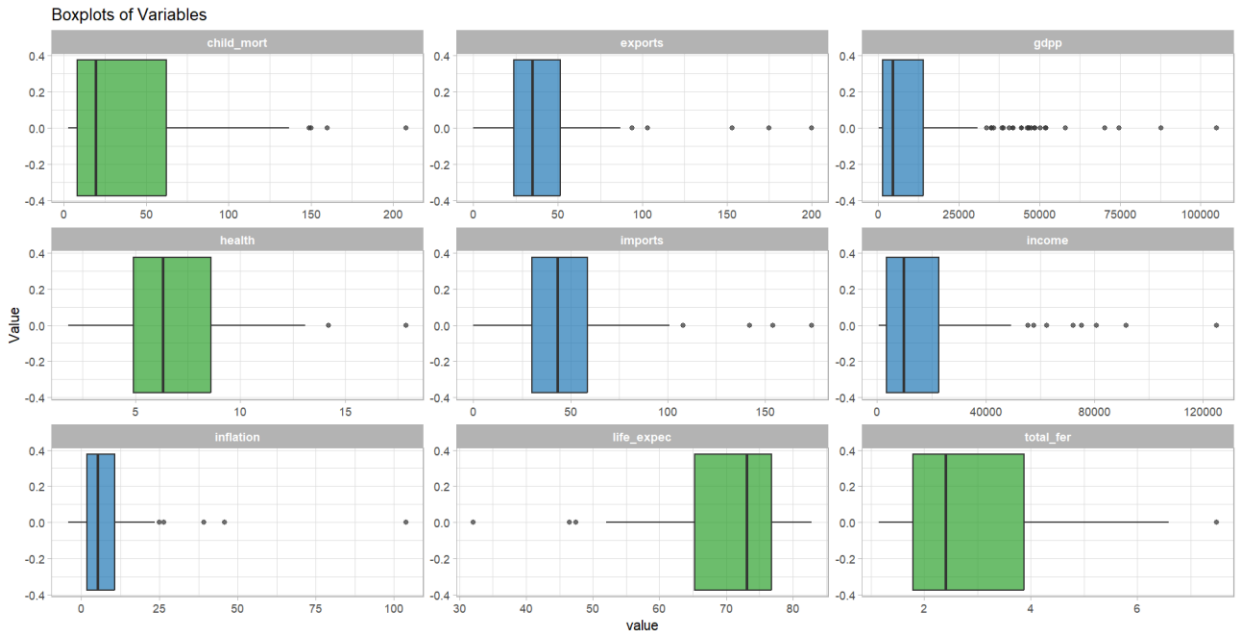


Figure 6: Boxplots showing outliers and spread of variables

Skewness and Correlation Analysis

Most of the variables in this dataset highly reflect the nature of inequality among countries. Looking first at the financial variables: GDPP, Income, and Inflation have very strong right skewness with many outliers, meaning only a small number of countries are rich. The differences among those wealthy outlier countries are clearly shown in the boxplots and affect the overall skewness. Imports and Exports also show signs of right skewness but are more evenly distributed with fewer outliers.

The health-related variables show both right and left skewness. Fertility Rate is right-skewed, meaning that most countries have fewer children, as the quality of healthcare is better, reflected by lower Child Mortality Rate (right-skewed) and longer Life Expectancy (left-skewed). Only Health Expenditure shows a more symmetric distribution with fewer outliers.

This dataset shows a strong correlation between economic development (financial variables) and health outcomes (health-related variables). Income and GDPP are highly correlated and are positively associated with Life Expectancy, while negatively associated with Child Mortality and Fertility Rates. This means that countries with higher Income and GDPP tend to have better healthcare and longer life expectancy, even with a decline in population growth. However, Health Expenditure has weaker correlations with most variables. Exports and Imports also show a moderate correlation to each other, meaning that countries that export more also import more.

```
> print(cor_matrix)
           child_mort    exports      health     imports     income  inflation  life_expec  total_fer       gdpp
child_mort  1.0000000 -0.3180932 -0.20040206 -0.12721092 -0.5243150  0.2882762 -0.88667610  0.8484781 -0.4830322
exports    -0.3180932  1.0000000 -0.11440840  0.73738083  0.5167836 -0.1072944  0.31631260 -0.3200106  0.4187248
health     -0.2004021 -0.1144084  1.00000000  0.09571668  0.1295786 -0.2553758  0.21069212 -0.1966740  0.3459655
imports    -0.1272109  0.7373808  0.09571668  1.00000000  0.1224062 -0.2469943  0.05439053 -0.1590484  0.1154982
income     -0.5243150  0.5167836  0.12957861  0.12240625  1.0000000 -0.1477560  0.61196247 -0.5018401  0.8955714
inflation   0.2882762 -0.1072944 -0.25537579 -0.24699428 -0.1477560  1.0000000 -0.23970496  0.3169211 -0.2216311
life_expec -0.8866761  0.3163126  0.21069212  0.05439053  0.6119625 -0.2397050  1.00000000 -0.7608747  0.6000891
total_fer   0.8484781 -0.3200106 -0.19667399 -0.15904843 -0.5018401  0.3169211 -0.76087469  1.0000000 -0.4549103
gdpp       -0.4830322  0.4187248  0.34596553  0.11549817  0.8955714 -0.2216311  0.60008913 -0.4549103  1.0000000
```

Figure 7: Correlation matrix of all variables

Part 2: Data Wrangling

Country and GDP were dropped from the clustering input, and the remaining variables were scaled using z-score normalization so that no single variable would dominate the distance calculations due to differences in scale.

```
> head(df_scaled)
  child_mort     exports      health     imports      income  inflation life_expec   total_fer
1  1.2876597 -1.13486665  0.27825140 -0.08220771 -0.80582187  0.1568645 -1.6142372  1.89717646
2 -0.5373329 -0.47822017 -0.09672528  0.07062429 -0.37424335 -0.3114109  0.6459238 -0.85739418
3 -0.2720146 -0.09882442 -0.96317624 -0.63983800 -0.22018227  0.7869076  0.6684130 -0.03828924
4  2.0017872  0.77305618 -1.44372888 -0.16481961 -0.58328920  1.3828944 -1.1756985  2.12176975
5 -0.6935483  0.16018613 -0.28603389  0.49607554  0.10142673 -0.5999442  0.7021467 -0.54032130
6 -0.5894047 -0.81019144  0.46756001 -1.27594958  0.08067776  1.2409928  0.5897009 -0.38178486
```

Figure 8: Data wrangling code and scaled data preview

```
> summary(df_scaled)
   child_mort        exports            health            imports            income          inflation        life_expec        total_fer       
 Min.   :-0.8845   Min.   :-1.4957   Min.   :-1.8223   Min.   :-1.9341   Min.   :-0.8577   Min.   :-1.1344   Min.   :-4.3242   Min.   :-1.1877  
 1st Qu.:-0.7444   1st Qu.:-0.6314   1st Qu.:-0.6901   1st Qu.:-0.6894   1st Qu.:-0.7153   1st Qu.:-0.5649   1st Qu.:-0.5910   1st Qu.:-0.7616  
 Median :-0.4704   Median :-0.2229   Median :-0.1805   Median :-0.1483   Median :-0.3727   Median :-0.2263   Median : 0.2861   Median :-0.3554  
 Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000  
 3rd Qu.: 0.5909   3rd Qu.: 0.3736   3rd Qu.: 0.6496   3rd Qu.: 0.4899   3rd Qu.: 0.2934   3rd Qu.: 0.2808   3rd Qu.: 0.7021   3rd Qu.: 0.6157  
 Max.   : 4.2086   Max.   : 5.7964   Max.   : 4.0353   Max.   : 5.2504   Max.   : 5.5947   Max.   : 9.1023   Max.   : 1.3768   Max.   : 3.0003  
```

Figure 9: Summary statistics of scaled variables

Part 3: Clustering Analysis

K-Means Clustering (PAM)

Optimal number of clusters: Both the Elbow Method and Silhouette Score plots suggest that the optimal number of clusters is 2.

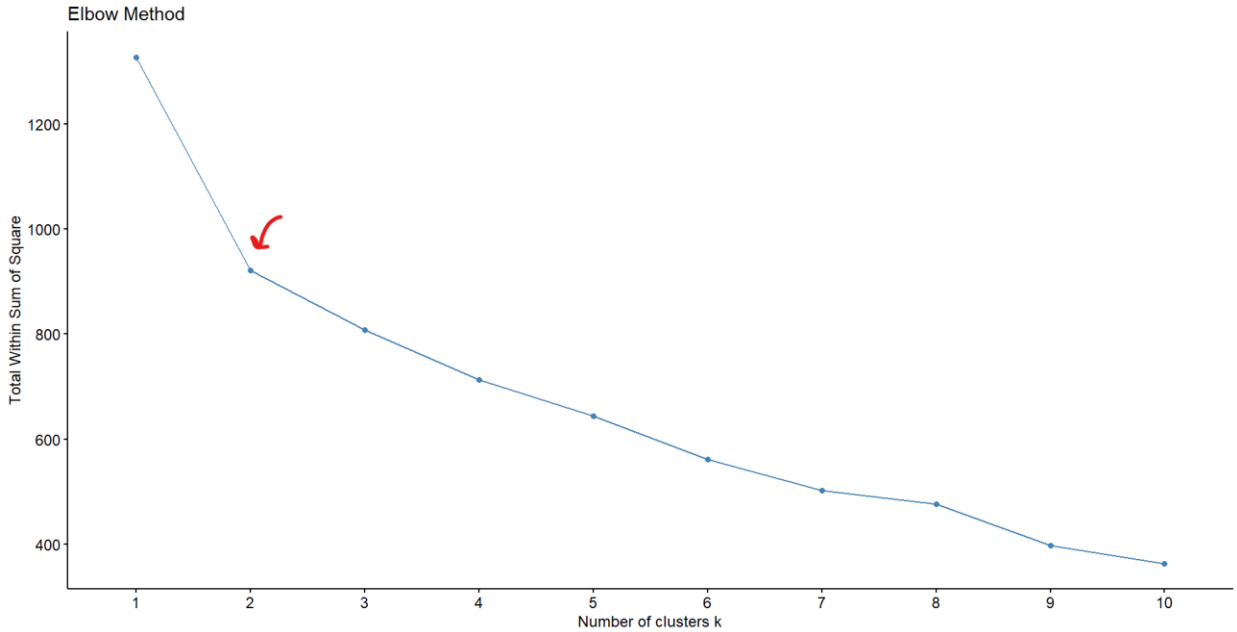


Figure 10: Elbow method for optimal cluster count

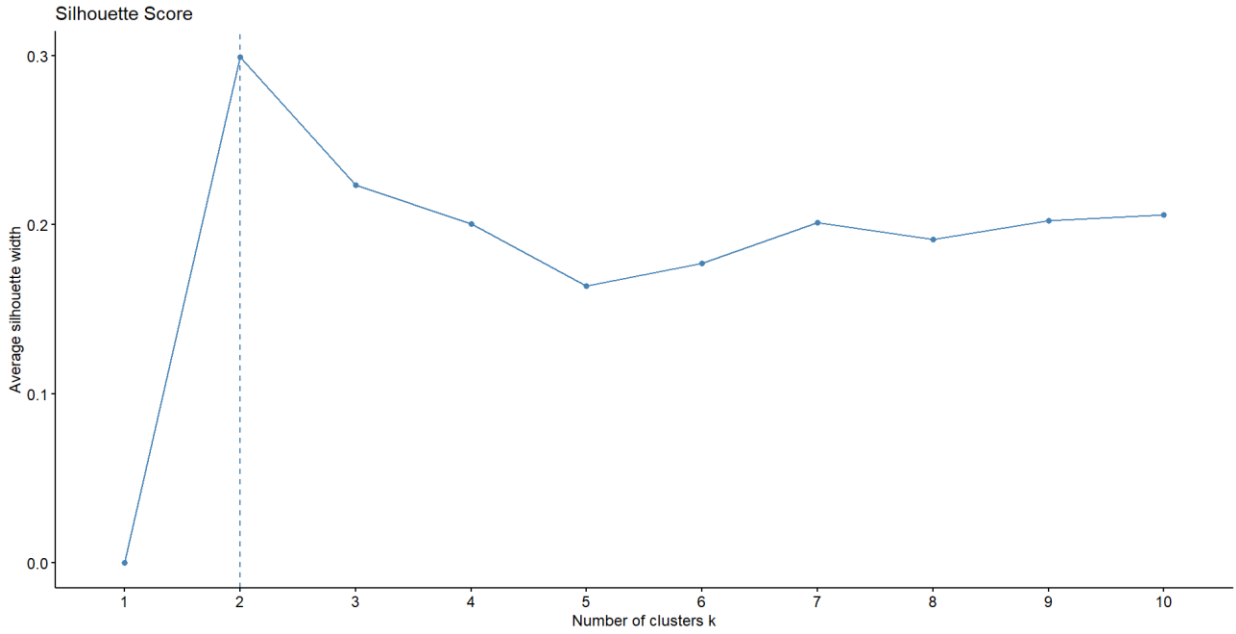


Figure 11: Silhouette score for optimal cluster count

Cluster labels were assigned and profiles analyzed using boxplots and scatterplots across all variables.

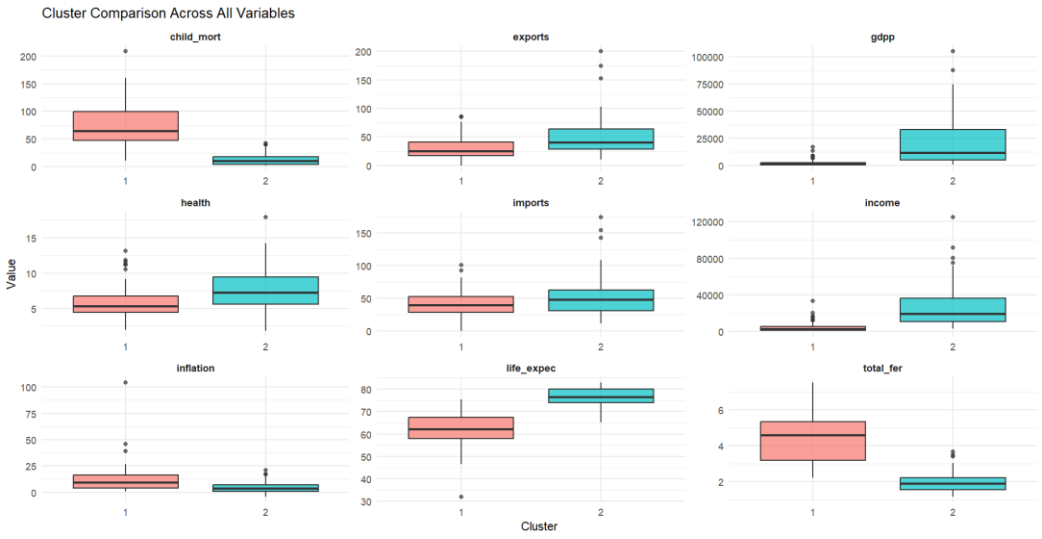


Figure 12: Cluster comparison across all variables (boxplots)

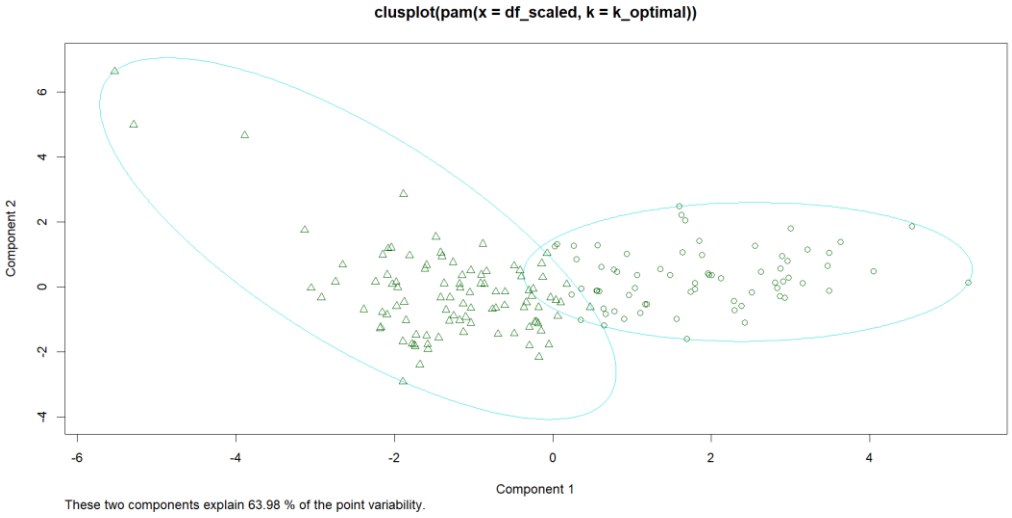


Figure 13: K-Means cluster scatterplot

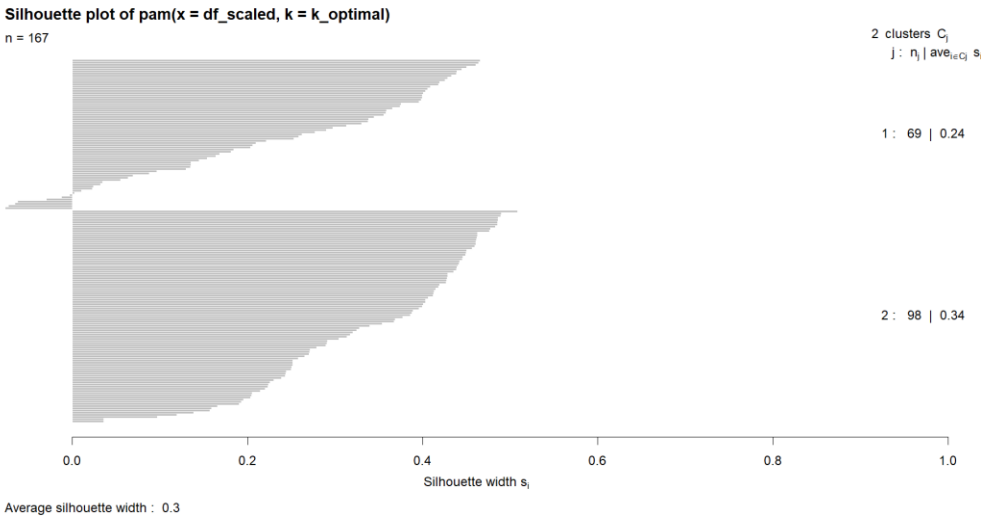


Figure 14: K-Means cluster visualization (silhouette/component view)

Hierarchical Clustering (AGNES)

Number of clusters: The potential number of clusters are 2 and 4, supported by the longest branches in the dendrogram and the gaps in the banner plot.

```
> print(aResult)
Call:    agnes(x = d, diss = TRUE, method = "ward")
Agglomerative coefficient:  0.9549308
Order of objects:
  [1]   1 156  26  65  27 127 133  18  29 151  37  57 130  81  94  32  33  98 113
 [20]  38  41 107  64  95 167 148  67  22 109 138 137  82  89 102  88   3  80  71
 [39] 141   6  72 126  23  34  36 161 154  35 120  46  63  47  13  70  20 121  48
 [58] 110 162  60  85 147  56 155  51 117 108   4 100  39  50  73 143 166 150 104
 [77] 164 114   2  49 125  62 106   7  77  31 142 128 152  10 144  15 157  17 119
 [96]  28  53 163  19  66  84   5 153  25  86 101  93  11  14  42 122  87 105  43
[115] 139  75  21 131  58  40 103  16 111 146  74  44 136  52  91  68 135  79  97
[134] 118 165  96 149 132   8  78  30  55  61 123  76 140 112 159   9  45  59  54
[153] 145  69 115 160  12  90 116 129 158  24  83 124  92 134  99
Height (summary):
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.2038  0.8218  1.3710  2.3648  2.2937 26.7781

Available components:
[1] "order"  "height" "ac"     "merge"  "diss"   "call"   "method"
```

Figure 15: AGNES agglomerative clustering output

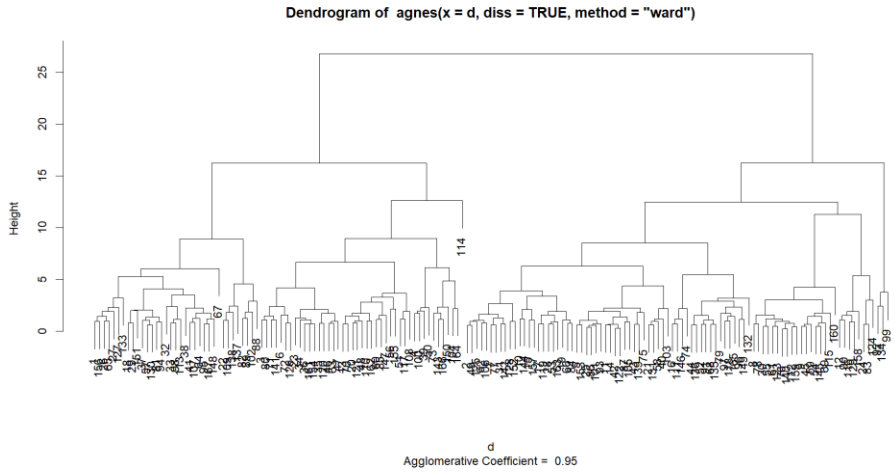


Figure 16: Dendrogram from AGNES clustering

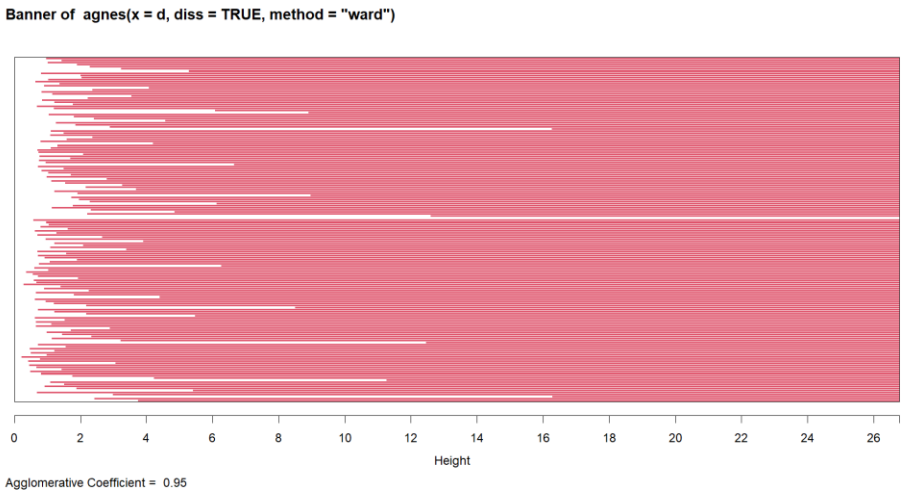


Figure 17: Banner plot from AGNES clustering

Comparison with K-Means: Using 2 clusters, the results from K-Means and Hierarchical methods are very similar. From the comparison table, most countries are assigned to the same clusters in both methods. Specifically, 63 countries in Cluster 1 and 83 countries in Cluster 2 are consistently grouped together, with only a small number of mismatches (6 and 15 countries). This indicates that the clustering structure is stable across methods.

```
> print(comparison_table)

   1  2
1 63  6
2 15 83
```

Figure 18: K-Means vs. Hierarchical cluster comparison table

Both Hierarchical and K-Means clustering produce very similar cluster profiles. In both methods, one cluster represents less developed countries with lower income and GDPP, higher child mortality and fertility rates, and lower life expectancy, while the other cluster represents more developed countries with higher income and GDPP, lower child mortality and fertility, and higher life expectancy. The summary statistics from both clustering methods show nearly identical patterns across all key variables.

```
> summary(subset(df, aClusters == 1))
   country           child_mort        exports           health          imports            income
 Length:78          Min.   :  8.70   Min.   : 0.109   Min.   : 1.970   Min.   :  0.0659   Min.   :  609
 Class :character   1st Qu.: 33.58   1st Qu.:18.750   1st Qu.: 4.487   1st Qu.: 26.6500   1st Qu.: 1678
 Mode  :character   Median : 62.65   Median :24.650   Median : 5.370   Median : 34.3500   Median : 3355
                    Mean   : 67.99   Mean   :28.553   Mean   : 6.144   Mean   : 38.0431   Mean   : 6411
                    3rd Qu.: 93.22   3rd Qu.:36.950   3rd Qu.: 7.845   3rd Qu.: 45.6500   3rd Qu.: 9778
                    Max.   :208.00   Max.   :85.800   Max.   :14.200   Max.   :101.0000   Max.   :33700
   inflation         life_expec      total_fer          gdpp        PAM_Cluster aClusters
 Min.   :  0.885   Min.   :32.10   Min.   :1.570   Min.   :  231   1:63        1:78
 1st Qu.:  4.968   1st Qu.:58.62   1st Qu.:2.603   1st Qu.:  672   2:15        2: 0
 Median :  8.855   Median :64.30   Median :4.160   Median : 1330
 Mean   : 12.075   Mean   :64.20   Mean   :4.038   Mean   : 3206
 3rd Qu.: 16.100   3rd Qu.:69.83   3rd Qu.:5.210   3rd Qu.: 4490
 Max.   :104.000   Max.   :79.10   Max.   :7.490   Max.   :17100
> summary(subset(df, aClusters == 2))
   country           child_mort       exports           health          imports           income
 Length:89          Min.   : 2.60   Min.   : 12.40   Min.   : 1.810   Min.   : 13.60   Min.   :  2520
 Class :character   1st Qu.: 4.50   1st Qu.: 32.70   1st Qu.: 5.380   1st Qu.: 38.40   1st Qu.:  9930
 Mode  :character   Median : 9.00   Median : 49.40   Median : 7.030   Median : 51.50   Median : 21100
                    Mean   :12.22   Mean   : 52.11   Mean   : 7.404   Mean   : 54.64   Mean   : 26551
                    3rd Qu.:17.20   3rd Qu.: 65.30   3rd Qu.: 9.480   3rd Qu.: 63.60   3rd Qu.: 38800
                    Max.   :44.40   Max.   :200.00   Max.   :17.900   Max.   :174.00   Max.   :125000
   inflation        life_expec      total_fer          gdpp        PAM_Cluster aClusters
 Min.   :-4.210   Min.   :63.00   Min.   :1.150   Min.   :   786   1: 6        1: 0
 1st Qu.: 0.873   1st Qu.:73.40   1st Qu.:1.510   1st Qu.:  4610   2:83        2:89
 Median : 2.590   Median :76.30   Median :1.870   Median : 12200
 Mean   : 4.019   Mean   :76.12   Mean   :1.993   Mean   : 21516
 3rd Qu.: 5.990   3rd Qu.:80.10   3rd Qu.:2.210   3rd Qu.: 35300
 Max.   :17.200   Max.   :82.80   Max.   :4.340   Max.   :105000
```

Figure 19: Hierarchical cluster profiles (summary statistics)

```
> summary(subset(df, PAM_Cluster == 1))
   country           child_mort       exports           health          imports           income
 Length:69          Min.   : 11.2   Min.   : 0.109   Min.   : 1.970   Min.   :  0.0659   Min.   :  609
 Class :character   1st Qu.: 47.0   1st Qu.:17.100   1st Qu.: 4.470   1st Qu.: 28.6000   1st Qu.: 1540
 Mode  :character   Median : 64.4   Median :25.000   Median : 5.250   Median : 39.2000   Median : 2660
                    Mean   : 74.8   Mean   :30.384   Mean   : 5.873   Mean   : 42.3865   Mean   : 4675
                    3rd Qu.: 99.7   3rd Qu.:41.100   3rd Qu.: 6.780   3rd Qu.: 52.7000   3rd Qu.: 5400
                    Max.   :208.0   Max.   :85.800   Max.   :13.100   Max.   :101.0000   Max.   :33700
   inflation         life_expec      total_fer          gdpp        PAM_Cluster aClusters
 Min.   :  0.885   Min.   :32.10   Min.   :2.200   Min.   :  231   1:69        1:63
 1st Qu.:  4.150   1st Qu.:57.90   1st Qu.:3.160   1st Qu.:  595   2: 0        2: 6
 Median :  8.920   Median :62.20   Median :4.560   Median : 1170
 Mean   : 12.087   Mean   :62.19   Mean   :4.356   Mean   : 2238
 3rd Qu.: 16.500   3rd Qu.:67.50   3rd Qu.:5.310   3rd Qu.: 2740
 Max.   :104.000   Max.   :75.40   Max.   :7.490   Max.   :17100
> summary(subset(df, PAM_Cluster == 2))
   country           child_mort       exports           health          imports           income
 Length:98          Min.   : 2.60   Min.   : 10.70   Min.   : 1.810   Min.   : 11.80   Min.   :  3340
 Class :character   1st Qu.: 4.65   1st Qu.: 28.43   1st Qu.: 5.585   1st Qu.: 31.32   1st Qu.: 10950
 Mode  :character   Median :10.30   Median : 39.95   Median : 7.210   Median : 47.20   Median : 19250
                    Mean   :12.55   Mean   : 48.66   Mean   : 7.480   Mean   : 50.06   Mean   : 25924
                    3rd Qu.:17.93   3rd Qu.: 63.58   3rd Qu.: 9.408   3rd Qu.: 62.58   3rd Qu.: 36200
                    Max.   :42.70   Max.   :200.00   Max.   :17.900   Max.   :174.00   Max.   :125000
   inflation        life_expec      total_fer          gdpp        PAM_Cluster aClusters
 Min.   :-4.210   Min.   :65.30   Min.   :1.150   Min.   :  1310   1: 0        1:15
 1st Qu.: 1.006   1st Qu.:73.92   1st Qu.:1.550   1st Qu.:  5162   2:98        2:83
 Median : 3.375   Median :76.40   Median :1.875   Median : 11600
 Mean   : 4.751   Mean   :76.44   Mean   :1.957   Mean   : 20516
 3rd Qu.: 7.152   3rd Qu.:79.97   3rd Qu.:2.192   3rd Qu.: 32975
 Max.   :20.900   Max.   :82.80   Max.   :3.660   Max.   :105000
```

Figure 20: K-Means cluster profiles (summary statistics)

Principal Component Analysis

PCA was applied to reduce dimensionality and visualize the clustering structure in two dimensions.

```
> summary(pca_res)
Importance of components:
                          PC1    PC2    PC3     PC4     PC5     PC6     PC7     PC8
Standard deviation     1.8907 1.2426 1.0786 0.85953 0.74980 0.47274 0.32945 0.29162
Proportion of Variance 0.4468 0.1930 0.1454 0.09235 0.07028 0.02794 0.01357 0.01063
Cumulative Proportion  0.4468 0.6398 0.7852 0.87759 0.94787 0.97580 0.98937 1.00000
```

Figure 21: PCA variance explained summary

```
> pca_res$rotation
                PC1         PC2         PC3         PC4         PC5         PC6         PC7         PC8
child_mort -0.4728799 -0.214124060 -0.09998804 0.115186553 -0.29716975  0.20332080  0.13513304  0.747903607
exports     0.3083961 -0.608374217  0.14603735 0.101508207 -0.05751086 -0.05344747  0.69641863 -0.109447681
health      0.1445682  0.241608166 -0.64740271 0.680155939  0.05895877  0.01392064  0.18267289 -0.044089348
imports     0.1946400 -0.661131277 -0.28525732 0.056360711  0.31536808 -0.03654309 -0.56924463  0.125062238
income      0.3867871 -0.031206520  0.24777586 0.315028674 -0.72825615  0.17896303 -0.35135794 -0.054302761
inflation  -0.2204750 -0.005770746  0.61577680 0.621291977  0.41786462  0.06357684 -0.08615010  0.009899599
life_expec  0.4641913  0.237343411  0.15808191 0.003856991  0.09136627 -0.60043527  0.02034424  0.577845597
total_fer  -0.4569516 -0.176701966 -0.05108475 0.159304266 -0.30353554 -0.74678145 -0.08968381 -0.272258407
```

Figure 22: PCA component loadings (rotation matrix)

```
> head(pca_df)
        PC1         PC2         PC3        PC4        PC5          PC6         PC7         PC8 PAM_Cluster
1 -2.89717025 -0.15783911 -0.90634204  0.3573477 -0.3978612 -0.2524293280 -0.45208003 -0.33928597           1
2  0.72190125  0.65421607 -0.11221647 -0.6177143  0.6654345  0.0780577947 -0.21516779  0.28748049           2
3 -0.09651877  0.47641205  1.35647114 -0.3165216  0.3898083 -0.4021710987  0.10929971  0.18623224           2
4 -2.99479252 -1.78253063  1.30721288  0.3267011 -0.5252342 -0.5434221244  0.51057059  0.24389752           1
5  1.17690487 -0.08360425 -0.06929212 -0.6543712  0.2400705 -0.2097668838 -0.23801278  0.07980457           2
6  0.05388965  1.77338643  0.89877855  0.8338569  0.4764980  0.0009697064  0.07883604 -0.07972731           2
```

Figure 23: PCA-transformed data preview

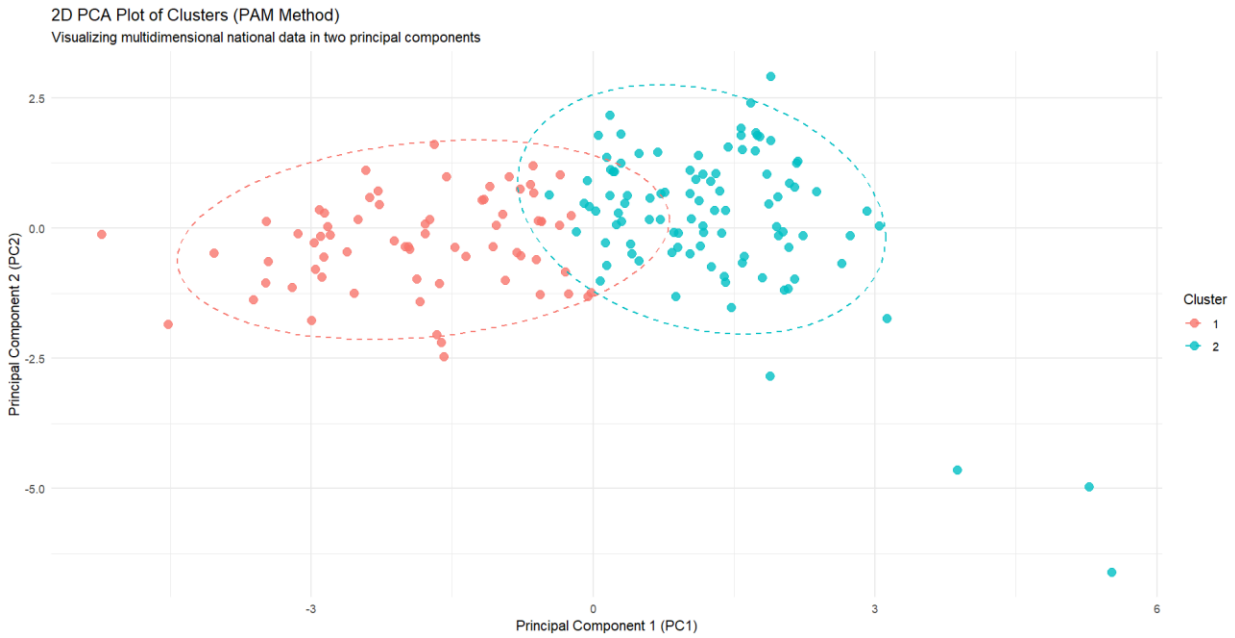


Figure 24: 2D PCA plot of clusters

Part 4: Interpretation and Reporting

Comparing Clustering Results Across Methods

The clustering results from K-Means, Hierarchical, and PCA are highly consistent. Both K-Means and Hierarchical clustering separate the countries into two clear groups, and the comparison table shows that most countries are assigned to the same clusters across methods with around 87% agreement.

The PCA visualization further supports this, as the clusters are clearly separated in the 2D space using the first two principal components, which explain about 64% of the total variance. The PCA visualization also confirmed that the clusters are well separated with a small degree of overlapping in reduced dimensional space, reinforcing the validity of the segmentation.

Socio-Economic Characteristics of Each Cluster

Cluster 1 (low-development group): This cluster is characterized by higher Child Mortality and Fertility Rates, but lower Life Expectancy. This is a common pattern among less developed countries. They lack access to quality healthcare and have poor nutrition and sanitation, while families tend to have more children either due to economic necessity or limited reproductive health resources.

Lower Income and GDPP also support this characteristic of less developed countries, since low Income and GDPP lead to limited healthcare access and poor living conditions that contribute to reduced life expectancy as well as limited imports and exports. Together, these indicators reinforce one another, forming a consistent pattern of structural underdevelopment.

Cluster 2 (high-development group): This cluster is characterized by lower Child Mortality and Fertility Rates, alongside higher Life Expectancy, which is a usual sign of more developed countries. These countries benefit from well-established healthcare systems, better nutrition, and effective sanitation infrastructure, while families tend to have fewer children due to higher education attainment, greater access to family planning, and overall higher standards of living.

Moreover, they also have higher Income and GDPP as stronger economic conditions enable sustained investment in healthcare, education, and public services.

Which Method Provided the Most Meaningful Segmentation

K-Means is the most suitable method for this analysis because it creates clear and easy-to-understand groups by minimizing differences within each cluster, which makes the country profiles more consistent and interpretable. Compared to hierarchical clustering, K-Means is more stable for this type of socio-economic data and works better when dealing with a larger number of countries, since it is less affected by early grouping decisions. This makes it more practical for separating countries based on indicators like income, child mortality, and life expectancy. PCA is useful for visualization and dimensionality reduction but it does not perform clustering itself. Instead, it supports K-Means by helping validate that clusters are meaningfully separated in reduced feature space.